World Wonders:

Exploring the Amazing wonders of the world with SQL

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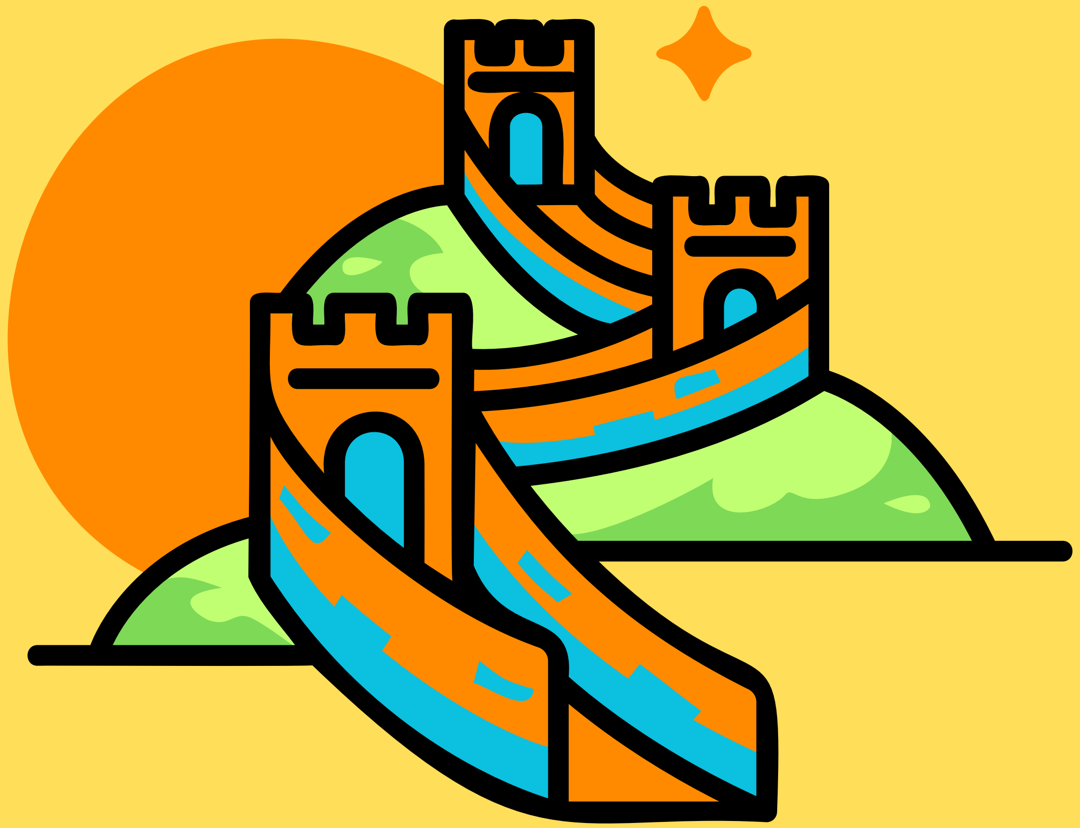
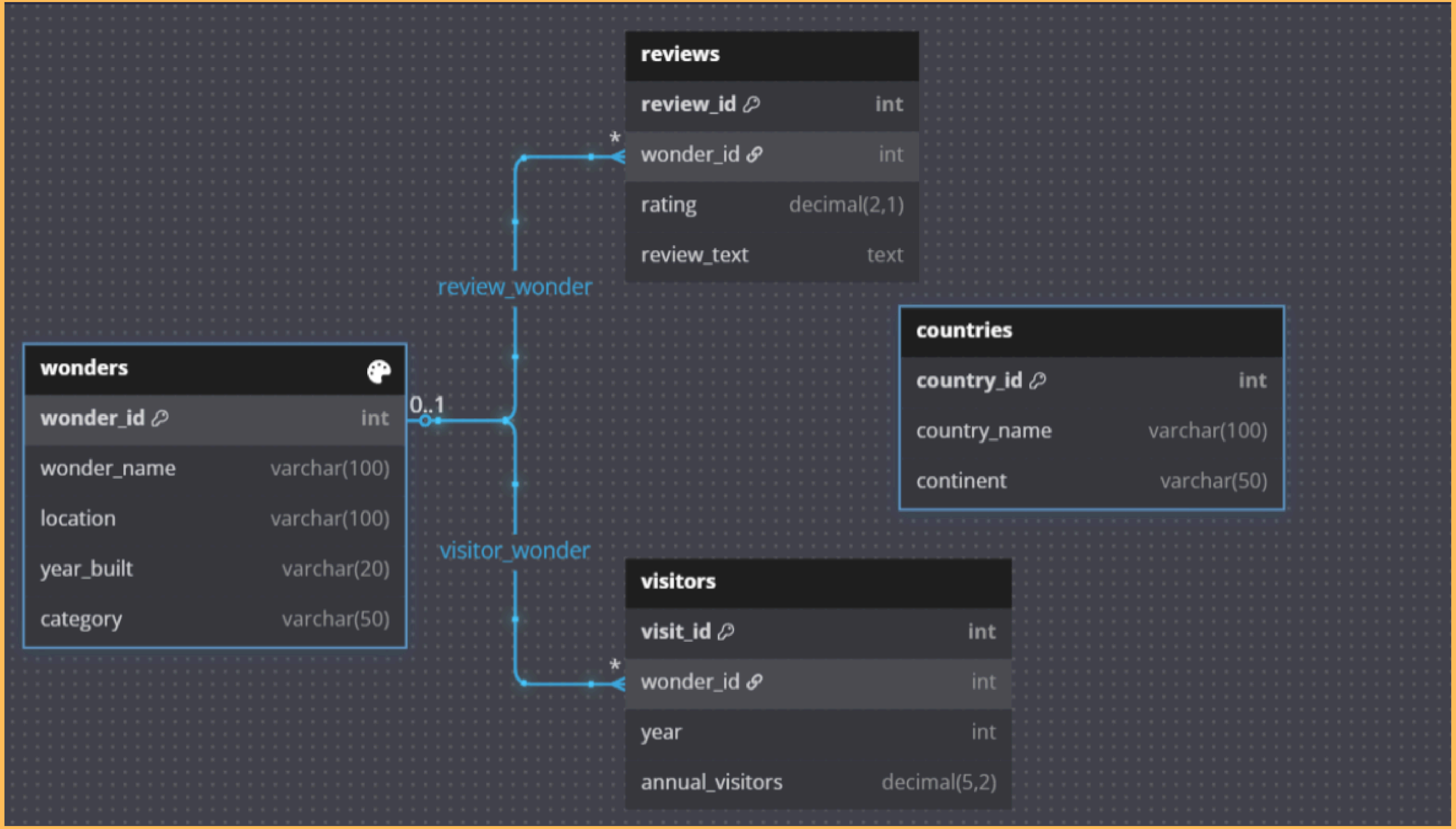


Overview

This project was a challenge given by Digits and Data, where I explored and analyzed data related to the Seven Wonders of the World. The goal was to extract meaningful insights by answering various analytical questions using SQL.

Schema


reviews
review_id int
wonder_id int
rating decimal(2,1)
review_text text
review_wonder
countries
country_id int
country_name varchar(100)
continent varchar(50)
wonders
wonder_id int
wonder_name varchar(100)
location varchar(100)
year_built varchar(20)
category varchar(50)
0..1
*
visitor_wonder
visitors
visit_id int
wonder_id int
year int
annual_visitors decimal(5,2)
*

Tables

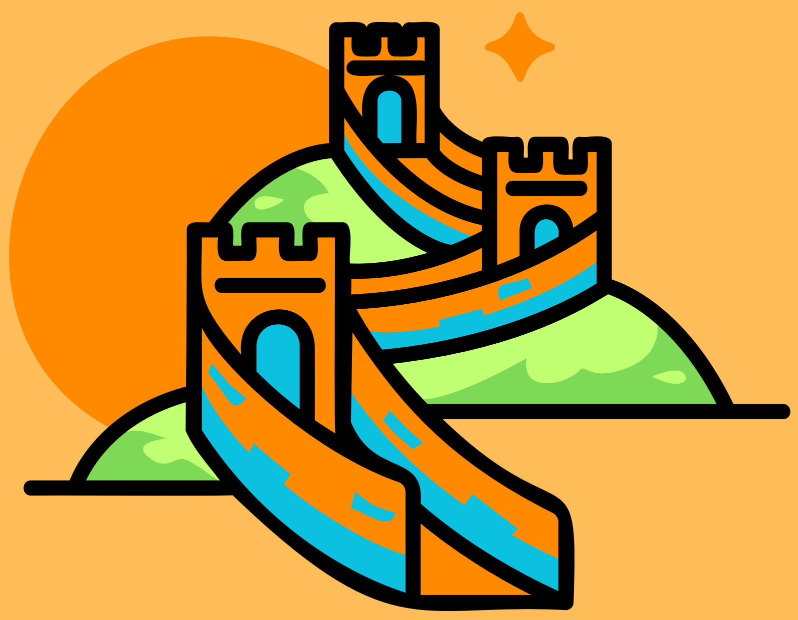
```
CREATE DATABASE IF Not Exists seven_wonders_db;

USE seven_wonders_db;

-- 1) Wonders Table (7 rows)
CREATE TABLE wonders (
    wonder_id INT PRIMARY KEY,
    wonder_name VARCHAR(100),
    location VARCHAR(100),
    year_built VARCHAR(20),
    category VARCHAR(50)
);
```

```
- 2) Visiters Table (20 rows)
CREATE TABLE visitors (
    visit_id INT PRIMARY KEY AUTO_INCREMENT,
    wonder_id INT,
    year INT,
    annual_visitors DECIMAL(5,2),
    FOREIGN KEY (wonder_id) REFERENCES wonders(wonder_id)
);
```

```
-- 3)Countries Table (7 rows)
CREATE TABLE countries (
    country_id INT PRIMARY KEY,
    country_name VARCHAR(100),
    continent VARCHAR(50)
);
```

```
-- 4)Reviews Table (15 rows)
CREATE TABLE reviews (
    review_id INT PRIMARY KEY AUTO_INCREMENT,
    wonder_id INT,
    rating DECIMAL(2,1),
    review_text TEXT,
    FOREIGN KEY (wonder_id) REFERENCES wonders(wonder_id)
);
```

1) Find all wonders with their location and category ?



```sql
-- 1) Find all wonders with their location and category ?

SELECT
        wonder_name ,
        location,
        category
FROM
      wonders;
```

Result Grid | Filter Rows: | Export:

wonder_name	location	category
Great Wall of China	China	Ancient
Christ the Redeemer	Brazil	Modern
Machu Picchu	Peru	Ancient
Chichen Itza	Mexico	Ancient
Roman Colosseum	Italy	Ancient
Taj Mahal	India	Medieval
Petra	Jordan	Ancient

2) Find the total visitors for each wonder in 2023, using 'CONCAT' function in annual_visitors column ?

```
SELECT
      w.wonder_name,
      CONCAT(SUM(v.annual_visitors)) AS Total_visitors_in_millions,
      v.year
FROM visitors v
    JOIN wonders w ON v.wonder_id = w.wonder_id
WHERE year = 2023
    GROUP BY  wonder_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

wonder_name	Total_visitors_in_millions	year
Great Wall of China	10.00	2023
Christ the Redeemer	2.00	2023
Machu Picchu	1.50	2023
Chichen Itza	2.60	2023
Roman Colosseum	7.60	2023
Taj Mahal	8.00	2023
Petra	1.00	2023

3) Find all visitor records for the Great Wall of China ?

```sql
-- 3) Find all visitor records for the Great Wall of China ?
SELECT w.wonder_name,
       v.annual_visitors as Total_visitors_in_million,
       v.year
       FROM visitors v
       JOIN wonders w ON v.wonder_id = w.wonder_id
WHERE wonder_name = 'Great Wall of China'
GROUP BY  v.year,
          Total_visitors_in_million;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

wonder_name	Total_visitors_in_million	year
Great Wall of China	10.00	2023
Great Wall of China	9.50	2022
Great Wall of China	8.70	2021

4) Count the total number of reviews in the reviews table ?

```sql
-- 4) Count the total number of reviews in the reviews table ?
SELECT
        COUNT(*) AS Total_reviews
FROM
        reviews;
```

Result Grid | Filter Rows:

Total_reviews
15

5) Find each wonder along with its continent ?

```sql
-- 5) Find each wonder along with its continent ?
SELECT w.wonder_name,
       c.continent
 FROM countries c
 JOIN wonders w ON c.country_name = w.location;
```

Result Grid | Filter Rows: | Export:

wonder_name	continent
Great Wall of China	Asia
Christ the Redeemer	South America
Machu Picchu	South America
Chichen Itza	North America
Roman Colosseum	Europe
Taj Mahal	Asia
Petra	Asia

6) Find the average annual visitors for each wonder from 2021 to 2023 ?

```sql
-- 6) Find the average annual visitors for each wonder from 2021 to 2023 ?
SELECT w.wonder_name,
      ROUND(AVG(v.annual_visitors),2) as avg_visitors_in_millions
FROM visitors v
JOIN wonders w ON v.wonder_id = w.wonder_id
WHERE
      year BETWEEN 2021 AND 2023
GROUP BY w.wonder_name;
```

Export: Wrap Cell

wonder_name	avg_visitors_in_millions
Great Wall of China	9.40
Christ the Redeemer	1.90
Machu Picchu	1.50
Chichen Itza	2.43
Roman Colosseum	7.20
Taj Mahal	7.77
Petra	0.95

7) List wonders along with their average rating ?

```sql
-- 7) List wonders along with their average rating ?

SELECT w.wonder_name,
    ROUND(AVG(r.rating),2) as Avg_rating
FROM reviews r
JOIN wonders w ON r.wonder_id = w.wonder_id
GROUP BY w.wonder_name;
```

Result Grid | Filter Rows: | Expo

wonder_name	Avg_rating
Great Wall of China	4.65
Christ the Redeemer	4.35
Machu Picchu	4.50
Chichen Itza	4.15
Roman Colosseum	4.55
Taj Mahal	4.85
Petra	4.07

8) Find wonders with visitor counts between 1 and 3 million in 2023 ?

```sql
-- 8) Find wonders with visitor counts between 1 and 3 million in 2023 ?
SELECT  w.wonder_name,
        v.annual_visitors AS visitors_count_in_million
FROM visitors v
JOIN wonders w ON v.wonder_id = w.wonder_id
where
      year = 2023
AND
   v.annual_visitors BETWEEN 1 AND 3
ORDER BY
         v.annual_visitors DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

wonder_name	visitors_count_in_million
Chichen Itza	2.60
Christ the Redeemer	2.00
Machu Picchu	1.50
Petra	1.00

9) Find reviews that contain the word ‘beautiful’ ?

```sql
-- 9) Find reviews that contain the word ‘beautiful’ ?

SELECT *
FROM reviews
WHERE review_text LIKE '%beautiful%' ;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

review_id	wonder_id	rating	review_text
4	2	4.2	Not as big as expected, but beautiful.

10) Find wonders located in either South America or Asia ?

```sql
-- 10) Find wonders located in either South America or Asia ?
SELECT  w.wonder_name,
        c.continent
FROM
       countries c
JOIN
       wonders w
ON
       c.country_name = w.location
WHERE
       c.continent IN ('South America', 'Asia');
```

Result Grid | Filter Rows: | Export:

wonder_name	continent
Great Wall of China	Asia
Christ the Redeemer	South America
Machu Picchu	South America
Taj Mahal	Asia
Petra	Asia

Conclusion

- This challenge helped reinforce my SQL proficiency while allowing me to analyze a real-world dataset effectively.
  It was a fun and learning challenge working on this project ,It also improved my problem-solving skills

Thank You So Much for your Attention

Pekamwar Gitanjali